

Matoshri Education Society's
MIT POLYTECHNIC AND ENGINEERING

Yeola

Department of Electrical Engineering

Vision

To develop skilled and competent professionals in Electrical Engineering who contribute to the technological advancement of society.

Mission

- To impart quality technical education through well-equipped laboratories and experienced faculty.
- To encourage students to develop practical skills and innovative thinking.
- To prepare students for successful careers in industry, research, and entrepreneurship.
- To promote ethical values and social responsibility among students.

Matoshri Education Society's
MIT POLYTECHNIC AND ENGINEERING

Yeola

Department of Electrical Engineering

DEPARTMENT OF ELECTRICAL ENGINEERING

Vision

To become a center of excellence in Electrical Engineering education, producing industry-ready professionals.

Mission

- To provide strong fundamental knowledge in electrical engineering principles.
- To develop problem-solving abilities and analytical skills.
- To foster research aptitude and encourage innovative project development.
- To build industry-academia partnerships for better employability.

A Project Report On

"OHM'S LAW"

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Under The Guidance Of:

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Department of Electrical Engineering
Academic Year: 2025-2026

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ANNEXURE A

CERTIFICATE

This is to certify that the project report entitled "OHM'S LAW" has been successfully completed by the following students of First Year Electrical Engineering:

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Under the guidance of Miss. Beldar G.N. at MIT Polytechnic and Engineering, Yeola during the academic year 2025-2026.

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INDEX

Sr. No	Contents	Page No
1	Abstract	1
2	Introduction	2
3	Statement of Ohm's Law	3
4	The Ohm's Law Formula ($V = IR$)	4
5	Water Analogy	5
6	V-I Characteristic Graph	6
7	Solved Examples	7
8	Practical Applications	8
9	Limitations of Ohm's Law	9
10	Conclusion	10
11	References	11

Abstract

Ohm's Law is one of the most fundamental principles in electrical engineering and physics. It establishes a linear relationship between voltage, current, and resistance in an electrical circuit.

This project report provides a comprehensive study of Ohm's Law, including its mathematical formulation ($V = IR$), its physical interpretation through the water analogy, practical applications in circuit design, and its limitations when applied to non-linear devices.

The report is intended for first-year engineering students to build a strong foundational understanding of electrical circuit analysis. Interactive simulations and solved numerical examples are included to reinforce the theoretical concepts.

The key objective of this project is to present Ohm's Law in an engaging and interactive manner, making it accessible to all students regardless of their prior background in physics or electronics.

Introduction

Electricity is an essential part of modern life, powering everything from household appliances to industrial machinery. Understanding the basic laws governing electrical circuits is crucial for any engineering student.

Ohm's Law, discovered by German physicist Georg Simon Ohm in 1827, is the cornerstone of circuit theory. It states that the current flowing through a conductor between two points is directly proportional to the voltage across the two points, provided the temperature remains constant.

Georg Simon Ohm published his findings in his book 'Die galvanische Kette, mathematisch bearbeitet' (The Galvanic Circuit Investigated Mathematically). Despite initial skepticism from the scientific community, his work was eventually recognized as a foundational contribution to electrical engineering.

This report explores Ohm's Law in depth, from its basic statement to its practical applications and limitations.



Statement of Ohm's Law

"The current flowing through a conductor is directly proportional to the potential difference (voltage) applied across its ends, provided the physical conditions such as temperature remain constant."

Mathematically, this is expressed as:

$$V = I \times R$$

Where:

- * V = Voltage (Potential Difference) measured in Volts (V)
- * I = Current measured in Amperes (A)
- * R = Resistance measured in Ohms (Ohm)

This relationship can be rearranged to find any of the three quantities:

- * $I = V / R$ (to find current)
- * $R = V / I$ (to find resistance)

The law applies to ohmic conductors, which exhibit a linear V-I characteristic.

The Ohm's Law Formula ($V = IR$)

The V-I-R Triangle

The Ohm's Law triangle is a visual tool used to remember the three forms of the formula. Simply cover the variable you want to find, and the remaining two variables show the formula.

Understanding Each Variable

- * Voltage (V): Think of it as the electrical 'pressure' that drives electrons through the circuit. A higher voltage means more force pushing the electrons.
- * Current (I): This is the rate of electron flow, similar to the flow rate of water in a pipe. More current means more electrons passing a point per second.
- * Resistance (R): This opposes current flow, like friction in a pipe. Higher resistance means less current for the same voltage.

Power Relationship

Combining Ohm's Law with the power formula $P = VI$ gives us additional useful relationships:

- * $P = I^2 \times R$ (Power in terms of current and resistance)
- * $P = V^2 / R$ (Power in terms of voltage and resistance)



Water Analogy

The water analogy is the most intuitive way to understand Ohm's Law for beginners.

Voltage = Water Pressure

Just as water pressure pushes water through pipes, voltage pushes electrons through a conductor. A higher water tank (more pressure) is analogous to a higher voltage battery.

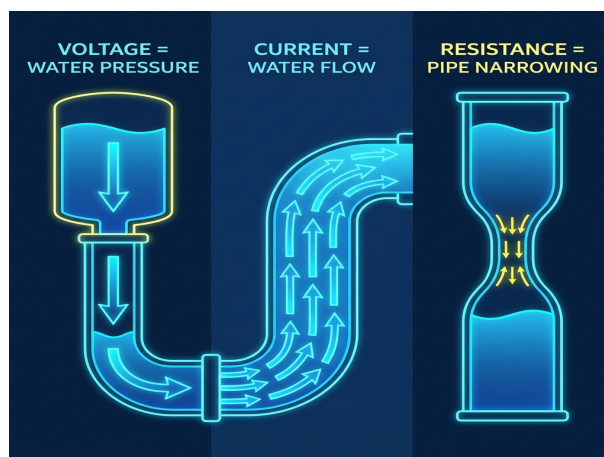
Current = Water Flow Rate

The amount of water flowing through a pipe per second is analogous to electrical current. More water flow equals more current.

Resistance = Pipe Constriction

A narrow section in a pipe restricts water flow, just as electrical resistance restricts current flow. A wider pipe allows more flow, equivalent to lower resistance.

This analogy helps students visualize abstract electrical concepts using everyday experience with water systems.



V-I Characteristic Graph

The V-I (Voltage-Current) characteristic graph is a fundamental tool in circuit analysis.

For Ohmic Conductors

When voltage is plotted on the X-axis and current on the Y-axis, an ohmic conductor produces a straight line passing through the origin. The slope of this line equals $1/R$.

Effect of Different Resistances

- * A low resistance (e.g., 10 Ohm) produces a steep line (high current for given voltage).
- * A medium resistance (e.g., 20 Ohm) produces a moderately sloped line.
- * A high resistance (e.g., 50 Ohm) produces a gentle slope (low current).

Key Observation

The linear nature of the V-I graph proves that the conductor follows Ohm's Law. Non-ohmic devices like diodes produce curved V-I characteristics.

Solved Examples

Example 1: Finding Current

Given: $V = 24V$, $R = 8 \text{ Ohm}$

Solution: Using $I = V/R = 24/8 = 3 \text{ Amperes}$

Example 2: Finding Voltage

Given: $I = 5A$, $R = 12 \text{ Ohm}$

Solution: Using $V = I \times R = 5 \times 12 = 60 \text{ Volts}$

Example 3: Finding Resistance

Given: $V = 120V$, $I = 0.5A$

Solution: Using $R = V/I = 120/0.5 = 240 \text{ Ohms}$

Example 4: Power Calculation

Given: $V = 230V$, $R = 100 \text{ Ohm}$

Solution: $I = V/R = 230/100 = 2.3A$; $P = V \times I = 230 \times 2.3 = 529 \text{ Watts}$

Practical Applications

Ohm's Law is applied extensively across electrical and electronics engineering.

1. LED Circuit Design

When connecting an LED to a power supply, Ohm's Law is used to calculate the required series resistor value to limit current and prevent the LED from burning out.

2. Power Supply Design

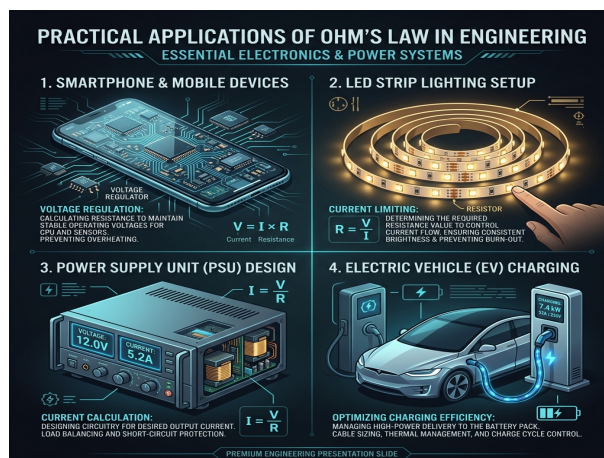
Engineers use Ohm's Law to determine current draw and power dissipation in voltage regulator circuits, ensuring safe operation of electronic devices.

3. Household Electrical Wiring

Electricians calculate wire gauge requirements and fuse ratings using Ohm's Law to ensure safe home electrical installations.

4. Troubleshooting Electrical Faults

Technicians diagnose faulty components by measuring voltage drops across circuit elements and comparing them to expected values using Ohm's Law.



Limitations of Ohm's Law

While Ohm's Law is fundamental, it has important limitations:

1. Non-Linear Devices

Diodes, transistors, and LEDs do not obey Ohm's Law. Their V-I relationship is non-linear and depends on the device characteristics.

2. Temperature Dependence

The resistance of most conductors changes with temperature. Ohm's Law assumes constant temperature, which is not always practical.

3. High-Frequency AC Circuits

In AC circuits with capacitors and inductors, impedance (not just resistance) must be considered. Ohm's Law in its basic form is insufficient.

4. Semiconductors

Materials like silicon and germanium have complex V-I characteristics that vary with doping concentration and temperature.

5. Superconductors

At extremely low temperatures, some materials exhibit zero resistance, making the concept of Ohm's Law inapplicable.

Conclusion

Ohm's Law ($V = IR$) is the most fundamental and widely used law in electrical engineering. Through this project, we have explored its definition, mathematical formulation, physical interpretation through the water analogy, and practical applications.

The key takeaways from this study are:

- * Voltage, current, and resistance are linearly related in ohmic conductors.
- * The V-I-R triangle provides a quick reference for solving circuit problems.
- * The water analogy makes abstract electrical concepts intuitive and accessible.
- * Ohm's Law has widespread applications from LED circuits to household wiring.
- * The law has limitations when applied to non-linear devices and non-standard conditions.

Understanding Ohm's Law is essential for any engineering student as it forms the basis for more advanced topics like Kirchhoff's Laws, Thevenin's Theorem, and AC circuit analysis.

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